

ADITYA KULKARNI

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EDUCATION

North Carolina State University

M.S. Computer Science

Raleigh, NC

Aug 2024 – May 2026

University of Mumbai

B.Tech. Computer Science, CGPA: 9.32/10

Mumbai, India

Aug 2020 – May 2024

WORK EXPERIENCE

Product Engineer Intern

May 2025 – Present

Isomer AI

Remote

- Shipped 10+ production backend features for an AI-powered automation platform using AI-assisted development (Claude, Cursor) – Temporal.io workflow orchestration, RBAC, real-time event pipelines – across 7 deployments in TypeScript, Node.js, PostgreSQL, and Redis
- Integrated multi-model LLM tooling across 3 providers (Claude, GPT-4o, Gemini) with MCP and built document processing pipelines (PDF generation, Microsoft Graph API, S3); contributed 394 commits and led code review in Agile sprint cycles

Software Developer Intern

Jun 2024 – Aug 2024

Mphasis (HP Company)

Remote

- Built 5+ AI-driven automation workflows including an NLP and ML-powered health claim processing pipeline, improving claim accuracy 25% in production; leveraged AI tools to accelerate feature delivery and reduce manual testing cycles
- Automated CI/CD deployments on AWS and GCP; integrated Grafana monitoring dashboards and conducted model validation to enhance production reliability

Graduate Research Assistant

Apr 2025 – Present

NC State University

Raleigh, NC

- Engineered a cloud-based genomic data platform on Google Cloud Run with JBrowse 2, enabling interactive visualization of 500GB datasets for 100+ researchers worldwide
- Improved ML model accuracy 12% through hyperparameter tuning and time-series analysis; collaborated on architecture design and model evaluation across cross-functional research teams

PERSONAL PROJECTS

LLM Eval Harness | Python, Anthropic/OpenAI/Gemini SDKs, DuckDB, Streamlit, Docker

2025

- Built a vendor-neutral LLM benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across grounded QA, multi-hop QA, and claim verification on a 2,000-example frozen test set with 3 repeated runs per model
- Tracked ExactMatch, TokenF1, hallucination rate, p50/p95 latency, and cost/call; applied paired bootstrap CIs and McNemar test for statistically rigorous model selection in production AI systems

Inventory Reservation Service | Java 21, Spring Boot 3.5, PostgreSQL, Testcontainers, Docker

2026

- Built a transactional inventory reservation REST API in Java 21 and Spring Boot preventing overselling via PostgreSQL row-level locking (SELECT FOR UPDATE) with idempotent endpoints and RFC 9457 error responses
- Verified concurrency correctness with Testcontainers – two concurrent requests against quantity 1 produce exactly one success and one 409 Conflict; Flyway migrations, Spring Security RBAC, multi-stage Docker, GitHub Actions CI

KubeServe – Kubernetes LLM Inference Platform | Python, FastAPI, Redis, Kubernetes, KEDA

2026

- Built a Kubernetes-native LLM inference platform: FastAPI router with 3-state circuit breaker and EWMA latency-aware load balancing across Claude, GPT-4o, and Gemini backends; KEDA autoscaler scales pods 0→N on queue depth
- Python Kopf CRD operator reconciles Deployments and KEDA ScaledObjects; OpenTelemetry + Prometheus observability; k6 load-tested to 500 VUs; 88 pytest tests; Helm + ArgoCD GitOps

TECHNICAL SKILLS

AI Tools: Claude Code, GitHub Copilot, Cursor, OpenAI API, LangChain, HuggingFace, MCP

Languages: Python, TypeScript, Java, JavaScript, SQL, R, C

AI/ML: PyTorch, TensorFlow, RAG, NLP, OCR, Pandas, NumPy, Prompt Engineering

Infrastructure: AWS, GCP, Docker, Kubernetes, PostgreSQL, Redis, Kafka, CI/CD, Grafana